

Question	Answer	Marks
4(b)(iv)	amplitude changes from maximum (at P) to minimum (at Q)	C1
	amplitude changes from maximum (at P) to minimum to maximum to minimum (at Q)	A1

Question	Answer	Marks
5(a)(i)	temperature decreases, resistance decreases	B1
5(a)(ii)	line from origin with decreasing gradient drawn in first quadrant or line from origin with decreasing gradient drawn in third quadrant	M1
	line drawn in the third or first quadrant of similar composition (straight line from origin followed by correct curve) and similar size as compared by eye to the first line	A1
5(b)(i)	$E = P / I$ $= 18 / 1.5$	C1
	$= 12 \text{ V}$	A1
5(b)(ii)	$I = 3.3 - 1.5$ $= 1.8$	C1
	$I = Anvq$ $1.8 = 1.4 \times 10^{-9} \times 3.4 \times 10^{28} \times v \times 1.6 \times 10^{-19}$	C1
	$v = 0.24 \text{ m s}^{-1}$	A1